use ambient RF signals for circuit operation and data transmission, it is usually not possible to control RF sources for transmitting power and operating on certain frequencies.

2. Ambient backscatter communicates ranging from several metres to tens of metres with low data rates. While it depends on scenarios where the technology is applied, if it needs to cover a large area, multiple devices are required.

3. The amount of electricity it collects is small. It can only collect enough electricity to power small sensors like LEDs, pressure sensors, accelerometers and so on. It can be faded due to noise.

4. Ambient backscatter may potentially face several security issues since backscatter transmitters are simple devices and RF sources are not controllable.

Applications

Development of ambient backscatter technology enables a method for device-to-device communication. It can be used for connecting the Internet of Things (IoT), which is the future of connected living. For example, a table could use this technology to alert someone who was around it and left a key on the table.

Smart sensors could be built, placed permanently inside any structure and set to communicate with each other to notify about the structure's condition.

Ambient backscatter can be adopted in many applications such as smart devices (wearables like health trackers to perform continuous analysis of a person's health condition), in logistics for tracking and medical equipment to monitor patient health even when there is no electricity and so on. This technology allows devices, that is, backscatter transmitters, to operate independently with minimum human intervention.

Some prototypes developed using this technology are: a smart card to process payments to another backscatter device used as a bus pass and a credit card-sized tracker used to track items and alert users when an item is misplaced.

In a nutshell

Ambient backscatter is a new communication technology that uses ambient RF signals to power devices and for communication. This technique does not require a dedicated power source—it does not generate its own RF signals. Due to this reason, it is more energy efficient than any conventional wireless communication.

In designing ambient backscatter, challenges relating to powering devices, controlling RF signals and protocols for transmission and reception were faced. Despite these, it proves to be a self-sustainable, low-cost communication system. This new battery-less communication system can revolutionise the world of low-power embedded devices and take a step towards the IoT. EFY